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16 October 1973
SUPERSEDING
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22 May 1953

MILITARY SPECIFICATION

BRIDGE ERECTION SET, FIXED BRIDGE,

BASIC SET, FOR BRIDGES, STEEL, RAILWAY

1. SCOPE

1.1 This specification covers the necessary equipment for the erection of through truss, unit construction, and I-beam railway bridges.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

T-R-605	- Rope, Manila and Sisal.
FF-B-575	- Bolt, Hexagon and Square.
FF-C-450	- Clamps, Wire Rope.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-T-276	- Thimbles, Rope.
FF-W-92	- Washers, Metal, Flat (Plain).
MM-L-736	- Lumber; Hardwood.
NN-K-231	- Kegs: Wood, Slack.
QQ-S-634	- Steel, Bar, Carbon, Cold Finished, (Standard Quality).
QQ-S-781	- Strapping, Steel, Flat and Seals.

FSC 5420

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QQ-S-790	- Steel Strapping, Round (Bare and Zinc-Coated).
QQ-T-830	- Tube, Steel, (Carbon, Mechanical; Seamless and Welded).
RR-C-271	- Chains and Attachments, Welded, Weldless, and Roller Chain.
RR-W-410	- Wire Rope and Strand.
TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
GGG-B-490	- Blocks, Tackle (Manila and Nylon Rope).
GGG-B-500	- Block, Tackle, (Wire Rope).
GGG-J-51	- Jacks, Mechanical, Hand Operated (Screw; Scissors; Reel; Planer; and Rack Bar).
GGG-J-1564	- Jacks, Dolly Type, Hydraulic.
GGG-W-636	- Wrenches (Box, Open End, and Combination).
GGG-W-641	- Wrench, Socket; (And Sockets, Handles, and Attachments for Socket Wrenches; Hand).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-H-904	- Hoists, Chain and Wire Rope, Hand Operated.
MIL-C-3131	- Cordage, Preparation for Delivery of.
MIL-H-3280	- Hoists, Chain, Manually Operated, Packaging of.
MIL-C-3774	- Crates, Wood; Open, 12,000- and 16,000- Pound Capacity.
MIL-B-3865	- Blocks, Rope, Tackle: Packaging of.
MIL-W-12065	- Winches, Drum, Hand-Operated; 2 or 3 Purchase.

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-163	- Steel Mill Products, Preparation for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

DRAWINGS

ME

D2664-1 thru- 34	- Launching Equipment for Railway Bridges.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36	- Structural Steel.
A108	- Cold-Finished Carbon Steel Bars and Shafting.
A120	- Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.
A576	- Special Quality Hot-Rolled Carbon Steel Bars.
B22	- Bronze Castings for Bridges and Turntables.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0 - Standard Qualification Procedure.
D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J492 - Rivets and Riveting.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, NY 10001.)

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3. REQUIREMENTS

3.1 Description. The bridge erection set or component(s), as specified (see 6.2) shall be fabricated as shown on Drawings D2664-1 through -34 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a bridge erection equipment set or component(s), as applicable, for examination within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods will produce erection equipment components that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Structural and rivet steel. Structural and rivet steel shall conform to ASTM A36.

3.3.2 Steel rods, bars, and shafting. Steel rods, bars, and shafting, except steel for the barrel drift pins, shall conform to ASTM A108 or QQ-S-634, grade designation 1030, cold rolled or cold drawn.

3.3.2.1 Steel for barrel drift pins. Steel for the barrel drift pins shall conform to ASTM A576, grade designation 1080, oil quenched-and-tempered to a Brinell hardness of 401 to 444.

3.3.3 Bronze. Bronze shall conform to ASTM B22, copper alloy No. 863.

3.3.4 Hardwood. Hardwood shall conform to MM-L-736, type II.

3.4 Steel tubing. Steel tubing shall conform to QQ-T-830, composition 1025, seamless, cold drawn and stress-relieved (finish annealed).

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3.5 Steel pipe. Steel pipe shall conform to ASTM A120, black, standard weight, except that the ends shall not be threaded and the hydrostatic test will not be required.

3.6 Bolts, nuts, and washers.

3.6.1 Bolts. Bolts shall conform to FF-B-575, grade 2 steel. Regular bolts shall be type II, unfinished, and turned bolts shall be type II, finished, with a tolerance for body diameter of plus or minus 0.008 inch.

3.6.2 Nuts. Nuts shall conform to FF-N-836, type II, style 4.

3.6.3 Threaded components. Threads for bolts, nuts, and pins, except where Acme threads are shown, shall be unified thread series, classes 2A and 2B; thread length shall be equal to 1-1/2 diameters except where otherwise specified on the drawings.

3.6.4 Washers. Washers shall conform to FF-W-92, type A, grade I, class A. Beveled washers shall be of malleable iron with one face beveled 2 inches in 12 inches; dimensions as follows: For 3/4-inch bolts - 11/32-inch average thickness, 13/16-inch diameter hole, 1-1/2 inches square; for 7/8-inch bolts -13/32-inch average thickness, 15/16-inch diameter hole, 2 inches square.

3.7 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification E6010, E6011, E6012, E6013, E6020, and E6030 as prescribed for the operation.

3.8 Chain hoists. Chain hoists shall conform to MIL-H-904, class 1 and as specified herein.

3.8.1 Spur-geared types. The 2-ton "Army type" hoists shall be type D, 12-foot lift, adjustable to an American Standard 12 I 31.8 beam. The 2-, 4-, and 10-ton hoists shall be type C, except that the lifts shall be 12-, 16-, and 16-foot respectively.

3.8.2 Racket lever type. The hoists shall be type H, with 10-foot lift.

3.9 Wrenches.

3.9.1 Socket wrench. The wrench shall conform to GGG-W-641, type I, class 2, 1-inch size, except that the length shall be approximately 6-1/2 inches, and the length of the handle approximately 8-1/2 inches.

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3.9.2 Structural wrenches. The wrenches shall conform to GGG-W-636, type IX, class optional, except that the wrenches for 1-inch bolts shall have a tang length of 3 inches and a maximum tang diameter of 5/8 inch.

3.10 Jacks. The 10-ton jack shall conform to GGG-J-1564 and the 25-, 50-, and 100-ton jacks shall conform to GGG-J-51 and as specified herein.

Capacity	Requirements
10-Ton	Type I.
25-Ton	Type I, class 1, style B, journal, 5 inch minimum rise.
50-Ton	Type I, class 1, style A, 14 inch minimum rise.
100-Ton	Type I, class 2, 12 inch minimum rise.

3.10.1 Traversing base for 50-ton jack. The base shall have a 4-inch height, with top plate dimensions 10 by 12 inches. An operating lever shall be provided. The base shall have a horizontal traverse of 15 inches carrying a fully loaded jack.

3.11 Manila rope blocks. The blocks shall conform to GGG-B-490, type II, shell length of 8 inches minimum, steel shell, triple sheave, loose side hook.

3.12 Wire rope blocks. The blocks shall conform to GGG-B-500.

3.12.1 Snatch blocks. Snatch blocks shall be type I.

3.12.2 Ordinary blocks. Ordinary blocks shall be type II, grade A, styles 2 and 3, 10-inch sheave for 5/8-inch rope size blocks, 12-inch sheave for 3/4-inch rope size blocks.

3.13 Sisal rope. Sisal rope shall conform to T-R-605, type S.

3.14 Wire rope. Wire rope shall conform to RR-W-410.

3.14.1 Guys. Guy rope shall be type I, class 2, 6 x 19, preformed, regular lay, equipped with spliced-in oval thimbles at both ends and one shackle. Thimbles shall be of a size to accommodate the shackle.

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3.14.2 Winch rope. Winch rope shall be type I, class 2, 6 x 19, preformed, regular lay.

3.14.3 Bridle slings. Bridle sling rope shall be type I, class 2, 6 x 19, preformed, regular lay. The bridle slings shall be formed from a single wire rope with a semi-enclosing type equalizing thimble in the center held in place by two zinc buttons. Each end of the rope shall be equipped with a spliced-in oval thimble and hook. The length indicated on the drawing is the length along one leg from the equalizing thimble to the center of the hook.

3.14.4 Equalizing sling. Equalizing sling rope shall be type I, class 3, 6 x 37, galvanized, preformed, regular lay. The sling shall be a two-part type, endless grommet equipped with two semi-enclosing type equalizing thimbles, one at each end of the loop. The length of the drawing is the distance between inside faces of thimbles.

3.15 Thimbles. Thimbles shall conform to FF-T-276, type III, pointed shape, size as applicable.

3.16 Wire rope clips. Wire rope clips shall conform to FF-C-450, type I, class 1.

3.17 Chain attachments. Attachments shall conform to RR-C-271.

3.17.1 Shackles. Shackles shall be type IV, class 4.

3.17.2 Links. Links shall be for type I, grade C, class 4 chain.

3.17.3 Hooks. Hooks shall be type V, class 3, capacity and size suitable for the application.

3.18 Winches. Winches shall conform to MIL-W-12065; heavy-duty type, capacity as specified on drawings.

3.19 Push car. The car shall be a 4-wheeled, 56-1/2-inch gage, 15-ton capacity railroad push car consisting of a detachable wood platform, 81 inches by 67 inches, mounted on a steel framework having the top surface of all members in the same horizontal plane. Wheels shall be of steel, 16 inches in diameter, equipped with roller bearings. Clearance between the platform and wheels shall be held to a minimum.

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3.20 Identification marking. Equipment, assemblies, and parts shall be identified in accordance with MIL-STD-130.

3.21 Treatment and painting. All parts of the equipment shall be treated and painted in accordance with MIL-T-704 with the following exceptions:

(a) Finish coat for members E159 to E191 inclusive shall be yellow No. 33538 lusterless.
(b) Finish coat for members E36N to E41 inclusive, E58N to E60AN inclusive, E122N to E129BN inclusive, E131N, E132N, and E133N shall be red No. 30252 lusterless.
(c) Finish coat for all remaining launching equipment, derricks, and miscellaneous equipment shall be gray No. 36231 lusterless.

In lieu of painting, all connecting bolts, nuts, threaded ends of pins, shafts and long bolts, drift pins, and wire rope and fittings shall be preserved in accordance with MIL-P-116, method I, type P-1.

3.22 Stenciling. The nomenclature, Federal Stock Number, net weight, cubage, and erection mark of each major component of the equipment shall be stenciled on each side of the component near the center. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. On small components, the stenciling shall be as large as practicable. Stenciling shall conform to MIL-T-704.

3.23 Workmanship. All parts, components, and assemblies of the bridge erection set shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material.

3.23.1 Fabrication. Steel used in the fabrication of the equipment shall be free from laminations, kinks, and sharp bends. Straightening of material shall be done by methods that will not cause injury to the metal. Corners shall be square and true. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Where re-entrant cuts cannot be made by shearing, a rectangular punch or router may be used. All burrs, rough edges, and sharp edges of holes shall be groundsmooth. Sawing, shearing, and chipping of steel used in fabrication shall be done neatly and accurately. Flame cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks.

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3.23.2 Bolt and pin connections. Boltholes shall be accurately punched or drilled and shall have all burrs removed. Steel jigs or templates shall be used for drilling or boring of all field-connected pin or boltholes. All bolts and nuts shall be tight.

3.23.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.23.4 Welders and welding.

3.23.4.1 Welders. Before assigning any welder or automatic welder equipment operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualification of the ASME Boiler and Pressure Vessel Code (Section IX).

3.23.4.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D1.1-Structural Welding Code". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.23.4 Tolerances. The position of all holes for field connections shall be within plus or minus 1/32 inch of their true position. The diameter of all drilled holes shall be within plus 0.015 inch and minus 0.005 inch of the diameter shown on the drawings. Pin and pin hole diameters shall be within plus or minus 1/64 inch. The tolerance for overall dimensions of members shall be within plus or minus 1/16 inch of those shown on the drawings. The tolerances specified for the

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positioning of holes and material and the length, width, and depth of member shall be referenced from one end or side of the member and shall be noncumulative.

3.23.5 Interchangeability. Like components shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable in any assembly of erection equipment sets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. The preproduction bridge erection set failing a performance test shall be disassembled in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

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4.3 Preproduction inspection.

4.3.1 Examination. The erection set or erection set component(s) as applicable shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests.

4.3.2.1 Assembly. The erection set shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection. When individual erection set components are procured separately, the test shall not apply.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Test. Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The erection set or erection set component shall be examined as specified herein, for the following defects:

MAJOR

101. Material not as specified.
102. Components missing or not as specified.
103. Workmanship not as specified.
104. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.

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105. Members not true to dimension and line, or containing twists or bends.
106. Rivets missing, misshapen, or incorrectly driven.

MINOR

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.
203. Stenciling not as specified.

4.5.2 Tests assembly. The supplier shall assemble one launching nose for the unit construction bridge complete with bracing for 12 foot 6 inch centers as shown on Drawing D2664-3; one launching nose for the through truss bridge complete with bracing and struts as shown on Drawing D2664-4; one tail frame complete with bracing as shown on Drawing D2664-14 and assembled on erection frames and trolleys as shown on Drawing D2664-34; and sufficient other component parts of the equipment to produce one complete working unit for inspection of fit, alinement, and operation. Interference of mating members or incorrect fit at joints shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107. Materials, methods and containers not as specified for level A. Each incorrect material, method, or container shall be considered one defect.
108. Unprotected surfaces not coated with preservative as specified for level A.
109. Jacks and traversing bases not preserved, retracted, and operating bases secured as specified for level A.

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110. Chain hoists not preserved, packaged and packed in accordance with the referenced document for level A.
111. Sisal rope not packaged and packed in accordance with the referenced document for level A.
112. Blocks not preserved, packaged, and packed in accordance with the referenced document for level A.
113. Wire rope and wire rope assemblies not preserved, packaged, and packed in accordance with the referenced document for level A.
114. Tools not preserved, packaged, and packed in accordance with the referenced document for level A.
115. Winches not preserved as specified for level A.
116. Push car components not preserved and packaged as specified for level A.
117. Loose bolts, nuts, and washers not preserved and packaged as specified for level A.
118. Miscellaneous components not preserved and packaged as specified for level A.
119. Consolidated packaging not as specified for level A.
120. Large bundled components not secured as specified for level A.
121. Small components not boxed or bundled as specified for level A.
122. Push cars not packed as specified for level A.
123. Marking missing, illegible, incorrect, or incomplete for level A or C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Unprotected surfaces. Unprotected exterior metal surfaces of components and equipment not specifically specified herein, and requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative.

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5.1.1.3 Jacks and traversing bases. The rams, rack bars, pawls, levers, and the threaded and unpainted surfaces of the jacks and traversing bases shall be coated with type P-1 preservative. The jacks shall be fully retracted. The operating levers shall be secured to the jack or base for which intended or packed together to prevent loss.

5.1.1.4 Chain hoists. The chain hoists shall be preserved and packaged in accordance with the level A requirements of MIL-H-3280.

5.1.1.5 Sisal rope. Each length of rope shall be packaged in accordance with the level A requirements of MIL-C-3131.

5.1.1.6 Blocks. Blocks shall be preserved and packaged in accordance with the level A requirements of MIL-B-3865.

5.1.1.7 Wire rope and wire rope assemblies. Wire rope and wire rope assemblies shall be preserved and packaged in accordance with MIL-STD-163.

5.1.1.8 Tools. Tools shall be preserved and packaged in accordance with the level A requirements of PPP-P-40.

5.1.1.9 Winches. Threaded surfaces of items such as adjusting rods, and bolts and nuts, also unpainted surfaces of items such as springs, brake handle, ratchet pawls, handcranks, pins, wire rope drum, grease fittings, gears, or any other metal surface which may become exposed by disassembly shall be coated with type P-1 preservative. Unpainted metal surfaces of the brake mechanism such as adjusting wedges, pins, cam lever, retracting springs, and the braking surface or face of the brake drum shall be coated with primer conforming to TT-P-664. Exterior components having threaded or other unpainted metal surfaces such as clevises, pins, etc., shall be coated with type P-1 preservative. Care shall be exercised to prevent preservative from coming in contact with the brake lining.

5.1.1.10 Push cars. When the wheels are removed from the car axles or spindles, the interior bearing surfaces of the hubs, the exterior bearing surfaces of the axle or spindle, and all bearings which are not removed from the axles or spindles shall be coated with type P-11 preservative. The coated surfaces of the axles or spindles and the bearings shall be wrapped with barrier material conforming to

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MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to PPP-T-60, type IV. Openings into the hubs shall be covered with the barrier material and tape specified herein. Bearings removed shall be individually preserved in accordance with MIL-P-116, method IC-1, applying type P-11 preservative.

5.1.1.11 Loose bolts, nuts, and washers. Loose bolts, nuts and washers shall be coated with type P-1 preservative. The bolts, nuts, and washers shall be packaged in quantities not to exceed 200 pounds gross weight in close-fitting boxes conforming to PPP-B-601, overseas type, grade B style optional; PPP-B-621, class 2, style optional or in wood kegs conforming to NN-K-231. The boxes or kegs shall not contain components of more than one size or description and shall be closed and strapped in accordance with the appendix to the applicable specification. When the quantity of bolts, nuts, and washers of like size and description are of such a small quantity or size to make use of wooden boxes or kegs impractical, they shall be packaged in close-fitting boxes conforming to PPP-B-636, W5c, style FOL, in quantities not to exceed the weight limitation of the box. The box shall be closed and sealed as specified in the appendix to the box specification for method V.

5.1.1.12 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.13 Miscellaneous components. Miscellaneous components not specifically identified herein requiring protection from corrosion or mechanical or physical damage shall be preserved and packaged as specified for components of similar design or construction.

5.1.1.14 Consolidated packaging. Packaged and unpackaged components requiring the protection afforded by a wood box, to accomplish the packing specified herein, shall be consolidated packaged together in close-fitting boxes conforming to PPP-B-601, overseas type, grade B, style optional or PPP-B-621, class 2, style optional. The contents shall be blocked and braced within the boxes to prevent movement or damage in accordance with MIL-STD-1186.

5.1.2 Level C. The components comprising a complete bridge erection set shall be preserved and packaged in a manner to provide protection against corrosion and physical or mechanical damage from the supplier to the initial destination.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Large components. Large components such as structural members and assemblies of the launching equipment including the derricks that do not require the protection of a shipping container shall be shipped loose or in secured bundles. Bundles shall not contain components of more than one size or description and shall be secured with strapping conforming to QQ-S-781, type I, finish B, size 1-1/4 by 0.035 inches. A minimum of two straps shall be used for lengths up to 6 feet, three straps for lengths up to 12 feet, and one additional strap for each additional 6 feet in length. Bundles shall be confined to a maximum of 6,000 pounds gross weight.

5.2.1.2 Small components. Small components and any items of equipment not specifically identified in 5.2.1.4, shall be packed in boxes or secured in bundles in the most convenient manner that will prevent loss or damage to the components and provide aid in segregation, erection and ease of handling. Bundled components shall be limited to a gross weight not to exceed 200 pounds. Boxed components shall not exceed the weight limitation of the box. To the extent practical only components of like description shall be contained within the same bundle or box. Bundled components shall be secured with heavy gage galvanized wire or with strapping conforming to QQ-S-790, finish 2, or QQ-S-781, finish B. Boxes shall conform to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, grade B, style optional. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.2.1.3 Push cars. The push cars shall be packed together in quantities of two each, in a crate conforming to MIL-C-3774, type I, style A. The one push car shall be placed up-side-down upon the other push cars and secured to prevent movement with strapping conforming to QQ-S-781, type I, finish B, size 1-1/4 by 0.035 inches. A minimum of four straps shall be used. Two straps shall be equally spaced from end-to-end around the platform and two other straps equally spaced from side-to-side around the platform. The push cars shall be secured to the crate base in a manner to prevent movement in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186.

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5.2.1.4 Equipment. Equipment preserved and packaged in accordance with basic specifications (see 5.1.1.4, 5.1.1.5, 5.1.1.6, 5.1.1.7, and 5.1.1.8) shall be packed in accordance with the level A requirements of the applicable specification.

5.2.1.5 Components procured individually. When individual components are procured they shall be packed as specified in 5.2.1.1, 5.2.1.2, 5.2.1.3, and 5.2.1.4, but only components of like description shall be boxed or bundled together.

5.2.2 Level C. The components comprising a complete bridge erection set or individually procured components shall be prepared for delivery as loose, bundled, or boxed components as applicable, and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules. Bundles and boxes shall contain only components of like description.

5.3 Marking. Loose components, bundles, packages, boxes and crates shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bridge erection set, when procured complete, contains erection equipment which allows simultaneous erection of two through truss, four unit construction, and four I-beam railway bridges, and also includes depot spares.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether complete bridge erection sets or component(s) are required (see 3.1).
(c) Time frame required for submission of the preproduction model (see 3.2).
(d) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production bridge erection equipment sets or components from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish sets or components conforming to this specification.

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6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each bridge erection equipment set.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of bridge erection equipment set.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 5420-A250

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-B-12745A(ME) Bridge Erection Set, Fixed Bridge, Basic Set, for Bridges, Steel, Railway

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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